

Vertical GaN Power Devices: Progress in Materials, Device Structures, and Breakdown-Field Management

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As the global demand for efficient energy conversion accelerates—driven by Artificial Intelligence (AI) data centers and Electric Vehicle (EV), electrification—Silicon (Si) power devices have approached their theoretical limits. Gallium Nitride (GaN) has emerged as a superior wide-bandgap (WBG) alternative. While Lateral GaN High-Electron-Mobility Transistors (HEMTs) dominate the low-voltage market, they suffer from surface-related reliability issues and poor area scaling at high voltages (>650 V). This paper presents an exhaustive review of Vertical GaN (v-GaN) technology. We analyze the fundamental physics favoring vertical architectures, including the decoupling of breakdown voltage from chip area and superior avalanche capability. Critical fabrication challenges—specifically epitaxial growth on native vs. foreign substrates, p-type doping via Mg-intercalated superlattices, and low-damage etching—are discussed in depth. We review state-of-the-art device topologies (CAVET, Trench MOSFET, FinFET) and advanced edge termination techniques (Bevel, JTE, FIT). Finally, we report on recent 2024-2025 breakthroughs, including the world's first vertical GaN-on-Si HEMT and hybrid GaN-on-SiC devices, providing a roadmap for v-GaN commercialization.

Index Terms—Vertical GaN, Wide Bandgap Semiconductors, Power MOSFET, CAVET, Edge Termination, Heteroepitaxy, Fluorine-Implanted Termination (FIT).

I. INTRODUCTION

THE demand for high-efficiency power conversion systems has driven Silicon (Si) devices to their theoretical limits. Wide Bandgap (WBG) semiconductors, particularly Gallium Nitride (GaN) and Silicon Carbide (SiC), offer significant improvements in power density and efficiency. GaN, with a critical electric field (E_c) of ≈ 3.3 MV/cm and high electron mobility, allows for devices with lower on-resistance (R_{on}) and faster switching speeds compared to Si.

Historically, the commercial GaN market has been dominated by Lateral GaN devices (GaN-on-Si HEMTs). However, as applications demand higher voltages (> 900 V) and higher power levels, the lateral architecture faces bottlenecks. Vertical GaN (V-GaN) devices, where the current flows through the bulk of

the drift layer, have recently gained significant research traction as the solution to these high-power challenges.

A. The Power Electronics Landscape

The performance and efficiency of modern power conversion systems are fundamentally constrained by the intrinsic properties of the semiconductor materials employed in their switching devices. Silicon (Si), despite its long-standing dominance in the semiconductor industry, is increasingly limited by its narrow bandgap energy of approximately 1.1 eV and relatively low critical electric field. These material constraints restrict its capability to operate reliably under high-voltage and high-temperature conditions, leading to substantial conduction and switching losses in advanced power systems.

Silicon Carbide (SiC) has emerged as a practical successor in high-voltage applications (900 V–3.3 kV),

enabled by its wider bandgap of 3.2 eV, superior critical electric field strength, and high thermal conductivity. These attributes have facilitated the commercialization of SiC MOSFETs and diodes in electric vehicles, renewable energy systems, and industrial motor drives.

Gallium Nitride (GaN), however, offers an even more favorable intrinsic material profile for next-generation power electronics. GaN exhibits a critical electric field on the order of 3.3 MV/cm—surpassing that of SiC—and an electron mobility approaching $2000 \text{ cm}^2/\text{V} \cdot \text{s}$, which is nearly three times higher than typical SiC drift-layer mobility. These combined material advantages yield a substantially enhanced Baliga Figure of Merit (BFOM), a metric that captures the fundamental trade-off between breakdown voltage (BV) and specific on-resistance ($R_{on,sp}$). When normalized to Si, the BFOM values for SiC and GaN are approximately 392 and 993, respectively, underscoring the theoretical potential of GaN for achieving simultaneously low conduction loss and high-voltage blocking capability.

Table I: Comparison of Different Substrates for GaN

Property	Sapphire	Si	SiC	GaN
Lattice Mismatch (%)	16	-17	3.1	0
Thermal Expansion				
Coefficient (10^{-6} K^{-1})	7.5	2.6	4.4	5.6
Cost	Low	Low	High	Very High
Thermal Conductivity				
($\text{W} \cdot \text{cm}^{-1} \cdot \text{K}^{-1}$)	0.25	1.5	3.0	2.1
Dislocation Density				
(cm^{-2})	High	High	Low	Very Low

In view of these intrinsic advantages, GaN is increasingly recognized as a strong candidate for high-efficiency power conversion systems, particularly where switching frequency, power density, and thermal performance are of primary concern.

B. Limitations of Lateral GaN Devices

Commercial GaN power devices are presently dominated by lateral high-electron-mobility transistors (HEMTs) fabricated on foreign substrates such as Silicon or Sapphire. While these architectures have enabled rapid market penetration in the sub-650 V regime,

they exhibit several intrinsic limitations that fundamentally restrict their scalability toward higher-voltage and high-reliability applications.

A primary constraint arises from the surface-dominated electric-field distribution inherent to lateral structures. Under high drain bias, the electric field exhibits pronounced crowding at the gate-drain access region, where the peak field resides almost entirely near the device surface. As a consequence, increasing the breakdown voltage (BV) requires extending the lateral gate-drain spacing or implementing complex field-plate structures, both of which lead to increased chip area, higher parasitic capacitances, and elevated fabrication cost. This lateral scaling requirement stands in stark contrast to vertical device architectures, in which the breakdown voltage is predominantly governed by the drift-layer thickness rather than device footprint.

Thermal management represents an additional bottleneck. Lateral GaN devices grown on foreign substrates rely on thick, low-thermal-conductivity buffer layers to mitigate lattice and thermal mismatch. These buffer stacks introduce substantial thermal resistance, impeding efficient heat extraction from the active channel. As switching frequencies and power densities increase—particularly in applications such as data-center power modules—the limited vertical heat flux in lateral GaN structures results in higher junction temperatures, degraded reliability, and constrained power-handling capability.

Device reliability is further challenged by the presence of surface states and trap-assisted charge modulation in the two-dimensional electron gas (2DEG) channel. These surface and interface traps can capture and release carriers during switching transients, leading to dynamic on-resistance (R_{on}) degradation and the well-known “current collapse” phenomenon. The susceptibility of the 2DEG to trapping effects also inhibits the ability of lateral GaN devices to exhibit avalanche behavior. Since the electric-field peak resides near the surface rather than within the bulk, lateral GaN HEMTs typically fail catastrophically during over-voltage events, limiting their suitability for applications

requiring avalanche ruggedness or fault tolerance.

Table II: Comparison between Lateral GaN and Vertical GaN

Feature	Lateral GaN	Vertical GaN
Substrate Differences	GaN-on-Silicon	GaN-on-GaN
Current Flow Paths	Horizontal	Vertical
Voltage Class	Up to 1700V	Over 1,200V
Current Density	Moderate	High
Size	Small	Smallest, compact system

Collectively, these limitations underscore the fundamental architectural constraints of lateral GaN devices and motivate the exploration of vertical GaN technologies that redistribute electric fields into the drift layer, improve thermal pathways, and enable avalanche-capable high-voltage operation.

C. The Case for Vertical GaN

Vertical GaN (v-GaN) power devices fundamentally differ from their lateral counterparts in that current is transported through the thickness of the GaN drift layer rather than along the surface. This architectural shift redistributes the electric field, modifies the breakdown characteristics, and enables significantly enhanced power-handling capability. Several intrinsic advantages arise from this vertical current-flow configuration, making v-GaN a promising candidate for high-voltage, high-density power applications.

Figure 2. The schematic structures of (a) a lateral GaN device and (b) a vertical GaN device.

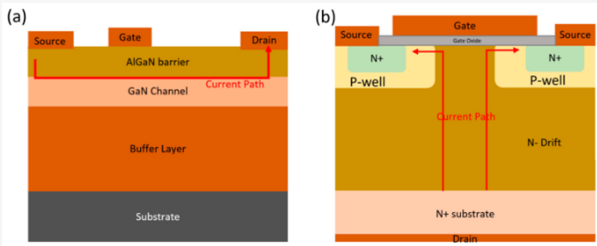


Fig. 1: lateral and vertical GaN

A principal benefit of the vertical structure is its favorable breakdown voltage (BV) scaling behavior. In contrast to lateral devices—where increasing BV necessitates enlarging the gate-drain spacing—the breakdown voltage in v-GaN is predominantly governed by the drift-layer thickness (d). To first order, the

relationship follows $V_{BD} \approx E_{crit} \cdot d$, where E_{crit} denotes the material's critical electric field. This decoupling of breakdown capability from lateral device area allows vertical devices to achieve high-voltage ratings while retaining compact footprints and reduced parasitic capacitances, yielding superior area efficiency compared to lateral GaN HEMTs.

Another key advantage is the improved avalanche robustness afforded by the bulk-centric electric-field distribution. By shifting the peak electric field away from the vulnerable surface region and into the uniformly doped drift epitaxy, v-GaN devices are capable of sustaining controlled avalanche events without catastrophic failure. This capability is essential for applications where fault tolerance, surge reliability, or transient overvoltage conditions are present. Such avalanche behavior is fundamentally inaccessible to lateral GaN devices, whose surface-dominated electric-field profiles typically result in destructive breakdown.

Furthermore, vertical GaN devices can support substantially higher current densities owing to their utilization of the full three-dimensional volume of the drift region for charge transport. The vertical conduction path reduces current crowding, enhances current spreading, and enables a larger effective cross-sectional area for conduction compared to the narrow lateral surface channel in HEMT structures. As a result, v-GaN devices achieve higher maximum current density (J_{max}) and improved thermal distribution, both of which are critical for high-power switching and compact module integration.

Collectively, these attributes—scalable breakdown voltage, inherent avalanche capability, and high current density—underscore the fundamental advantages of the vertical GaN architecture and motivate its development as a next-generation platform for high-voltage, high-efficiency power electronics.

II. MATERIAL GROWTH AND SUBSTRATE CHALLENGES

The fabrication of v-GaN devices is fundamentally limited by the availability and quality of substrates.

A. Homoepitaxy (GaN-on-GaN)

From a materials science perspective, growing Gallium Nitride (GaN) devices on native GaN substrates represents the optimal approach for realizing high-performance vertical power electronics.¹ Unlike heteroepitaxy on foreign substrates, homoepitaxial growth eliminates the fundamental challenges of lattice mismatch and thermal expansion coefficients mismatch. This structural coherency allows for the deposition of drift layers with exceptional crystalline quality, achieving dislocation densities as low as $10^3 - 10^5 \text{ cm}^{-2}$. Such a reduction in defect density is critical for vertical devices, as it directly correlates with minimized leakage currents and a significantly higher resistance to premature breakdown mechanisms.

GaN-on-GaN Offers High Ruggedness

Vertical construction of GaN-on-GaN offers innate robustness based on its homoepitaxial structure, compared to lateral devices such as GaN-on-Si / GaN-on-SiC

Attribute	GaN-on-Si	GaN-on-SiC	GaN-on-GaN	Impact
Defect Density (cm^{-2})	10^9	5×10^8	10^3 to 10^5	Better Yields, Higher Reliability
Lattice Mismatch (%)	17	3.5	0	Elimination High stress in Epitaxy causing Dislocations → Better Reliability
CTE Mismatch %	54	25	0	No Wafer deformations (cracking, warping)
Layer Thickness μm	1-2	2-6	>40	Higher BV → Higher voltage applications

Fig. 2: GaN on GaN

In terms of substrate fabrication, the industry currently relies on specific growth methodologies to balance quality and throughput. Hydride Vapor Phase Epitaxy (HVPE) has established itself as the industry standard for producing thick, free-standing GaN substrates, favored for its high throughput with growth rates ranging from 50 to 100 $\mu\text{m}/\text{h}$. Alternatively, the Na-flux method (High Nitrogen Pressure Solution Growth) is utilized to produce crystals of even higher quality. However, this technique presents significant engineering hurdles, requiring sophisticated high-pressure equipment capable of sustaining environments up to 20 kbar, which limits its ease of scalability compared to vapor phase methods.

Despite the clear technical superiority of homoepitaxial GaN, widespread commercial adoption is currently constrained by economic factors. The production

cost of native GaN substrates remains the primary barrier, significantly exceeding that of mature substrate technologies such as Silicon (Si) or Silicon Carbide (SiC). Consequently, while the performance benefits are indisputable, bridging the cost gap through improved manufacturing yields and larger wafer diameters remains the central challenge for the next generation of GaN power devices.

B. Heteroepitaxy (GaN-on-Foreign Substrates)

To mitigate costs, researchers employ heteroepitaxy on foreign substrates, though this compromises crystal quality. GaN-on-Sapphire suffers from a 16% lattice mismatch and poor thermal conductivity, while GaN-on-Si faces a -17% mismatch and thermal expansion disparities that induce wafer cracking. These structural imperfections result in dislocation densities exceeding 10^8 cm^{-2} , causing leakage currents orders of magnitude higher than native substrates. This degradation is dominated by Variable-Range Hopping (VRH) conduction through the dense defect network.

III. CRITICAL PROCESS TECHNOLOGIES

Successful v-GaN fabrication requires mastering three key processes: Epitaxy, Doping, and Etching.

A. P-Type Doping and Mg Activation

The realization of highly conductive p-GaN remains a fundamental bottleneck in GaN technology due to the intrinsic limitations of Magnesium (Mg) as an acceptor. Mg is characterized by a high activation energy, which results in poor ionization efficiency at room temperature.

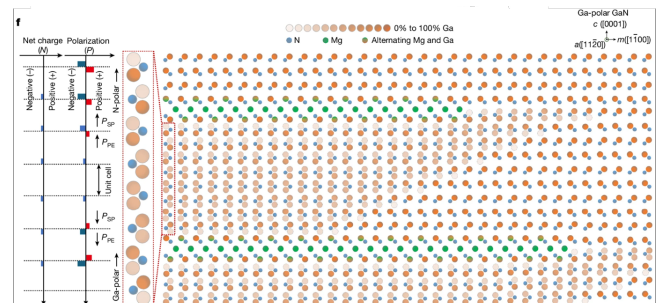


Fig. 3: MiGs

Furthermore, aggressive doping strategies typically trigger self-compensation mechanisms, where the formation of defects, such as nitrogen vacancies, counteracts the desired hole concentration.

In a significant 2024 advancement, Wang et al. addressed these constraints by developing 2D-magnesium-intercalated GaN superlattices (MiGs). By annealing Mg thin films to induce interstitial diffusion into monolayer atomic sheets, this technique successfully decouples carrier mobility from dopant concentration. This breakthrough allows for the synthesis of p-GaN with concurrently high hole density and mobility, offering a critical solution for minimizing contact resistance in vertical GaN devices.

B. Advanced Etching Techniques

Fabricating vertical architectures like Trench MOSFETs and FinFETs necessitates deep etching, typically achieved via Inductively Coupled Plasma (ICP) using Cl_2/BCl_3 . However, this process invariably introduces plasma-induced sidewall damage, including silicon impurities and nitrogen vacancies. To mitigate these defects, a hybrid etching strategy is employed, where ICP is followed by wet etching with hot Tetramethylammonium Hydroxide (TMAH). This post-treatment selectively removes the damaged layer to reduce surface roughness (R_q) and aligns the trench to the m- or a-plane, thereby optimizing channel carrier mobility.

IV. DEVICE ARCHITECTURES

Several V-GaN topologies are currently under intense research and early commercialization.

A. Current Aperture Vertical Electron Transistor (CAVET)

The CAVET combines the high mobility of the AlGaIn/GaN channel (similar to HEMT) with a vertical drift region. Current flows through an aperture into the bulk. This structure offers low channel resistance but involves complex processing to create the current blocking layer (CBL).

The CAVET combines the high channel mobility of an AlGaIn/GaN heterojunction (like a HEMT) with a vertical drift region.

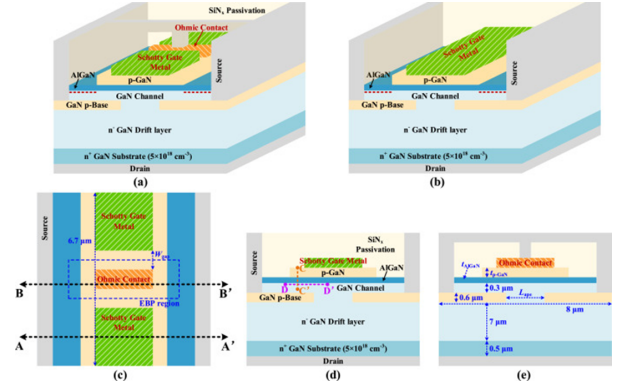


Fig. 4: CAVET

The CAVET architecture functions by directing current through a surface 2DEG channel and an aperture within a Current Blocking Layer (CBL) to the drain. This design advantageously utilizes polarization-induced 2DEG to minimize on-resistance without the need for p-type channel doping. However, its practical realization is significantly hindered by the complexity of fabricating the CBL, as the required epitaxial regrowth or ion implantation steps frequently introduce performance-limiting interface impurities.

B. Trench MOSFETs

Trench MOSFETs, adapted from mature Si and SiC technologies, represent the most promising topology for realizing normally-off GaN power devices. The architecture operates by etching a gate trench through the n+ source and p-body layers into the drift region, inducing a conductive channel along the vertical sidewall. While deep trench etching has historically challenged GaN fabrication due to defect generation, recent strides in low-damage etching and dielectric deposition have enabled the production of devices with stable threshold voltages ($V_{th} > 3V$) and breakdown voltages exceeding 1200 V.

The primary figure of merit for these devices involves minimizing specific on-resistance ($R_{on,sp}$) while maintaining robust blocking capabilities. However, reliability remains contingent on the quality of the gate oxide dielectric, particularly at trench corners where

high electric field crowding is prone to induce dielectric breakdown. To mitigate this, corner rounding via wet etching techniques is critical to alleviate field stress and ensure long-term device stability.

C. Vertical FinFETs

FinFETs utilize narrow semiconductor "fins" wrapped by the gate. This provides superior electrostatic control, suppressing short-channel effects and lowering leakage current. Recent demonstrations include optically triggered FinFETs, which isolate the gate driver from high-voltage noise.

V. EDGE TERMINATION TECHNOLOGIES

In vertical power devices, the electric field tends to crowd at the edge of the active area, leading to premature breakdown. Effective edge termination is required to spread this field and approach the theoretical breakdown limit of the material.

A. Bevel Mesa Termination

Bevel termination modulates the electric field at the device periphery by etching the edge at a specific angle (θ). This technique is categorized into positive or negative bevels, depending on whether the device's cross-sectional area decreases or increases, respectively, as it transitions from the heavily doped to the lightly doped region.

Physically, employing a shallow bevel angle effectively stretches the surface depletion region, thereby reducing the peak surface electric field (E_{peak}) to prevent premature breakdown. However, the practical application of this method is often limited by manufacturing complexities, particularly the difficulty in precisely controlling the acute angles ($< 10^\circ$) required for optimal field relief.

B. Junction Termination Extension (JTE)

JTE involves creating zones of varying p-type charge (P_1, P_2, P_3) at the edge. By grading the charge, the electric field is gradually reduced³⁵. Simulation results show that optimizing the JTE dose and length can effectively flatten the electric field profile, preventing crowding.

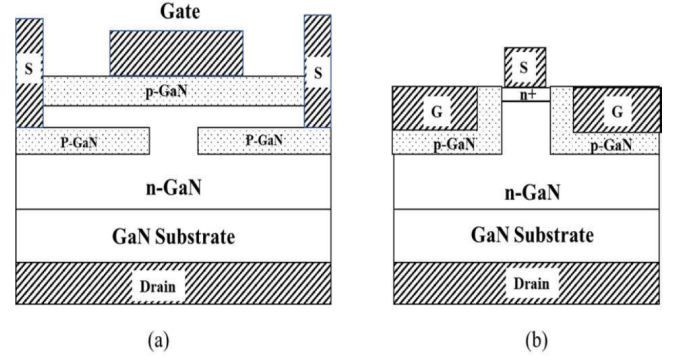


Fig. 5: JFETs

C. Guard Rings (GR)

Floating p-type rings are placed around the main junction. When the depletion region reaches a ring, it "punches through," allowing the ring to share the potential drop. Guard rings are highly robust but require precise spacing design.

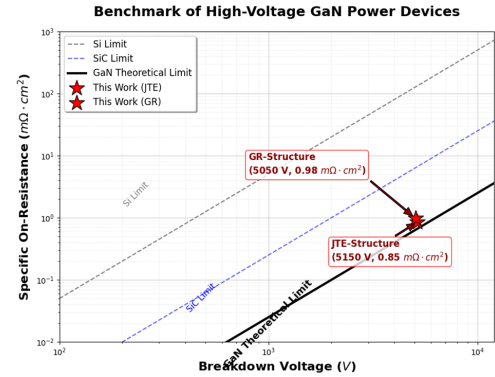


Fig. 6: Benchmark of High-Voltage GaN Power Devices

D. Fluorine-Implanted Termination (FIT) - 2025 Breakthrough

In a significant advancement for vertical GaN technology, a collaboration between Huawei and Shandong University has introduced the Fluorine-Implanted Termination MOSFET (FIT-MOS). This novel architecture addresses the critical challenge of edge termination in power devices, aiming to enhance breakdown robustness while simplifying the fabrication process compared to conventional methodologies.

The fundamental mechanism of the FIT-MOS relies on the strategic implantation of fluorine ions into the GaN drift layer. These implanted ions function

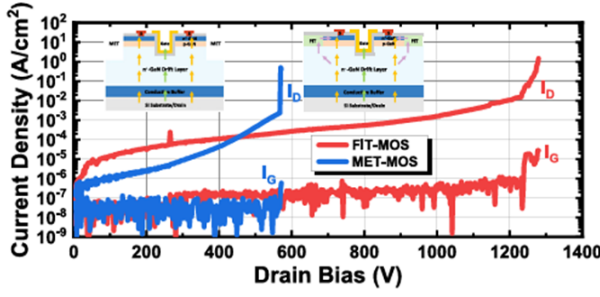


Fig. 7: FIT

as fixed negative charges that actively modulate the electric field distribution within the device. By artificially expanding the depletion region, the design effectively shields vulnerable high-field areas from premature breakdown. Crucially, this technique eliminates the need for complex epitaxial p-GaN regrowth, a process often associated with higher manufacturing costs and technical variability.

Experimental characterization validates the efficacy of this approach. The FIT-MOS demonstrated a substantial improvement in blocking capability, achieving a breakdown voltage of 1277 V. This represents a more than twofold increase compared to the 567 V observed in standard mesa-etched termination (MET) devices. Furthermore, this enhancement in voltage rating was achieved without compromising conduction efficiency, as the device maintained a low specific on-resistance ($R_{on,sp}$) of $5.6 \text{ m}\Omega \cdot \text{cm}^2$, indicating a superior trade-off between breakdown voltage and on-state resistance.

VI. RECENT BREAKTHROUGHS (2024-2025)

The period of 2024-2025 has seen transformative advances in v-GaN, pushing it closer to commercialization.

A. World's First Vertical GaN HEMT on Silicon (Ziener)

In a landmark development for power semiconductor integration, Dr. Li Zilan and the research team at Guangdong Ziener have successfully fabricated a vertical GaN High-Electron-Mobility Transistor (HEMT) on a Silicon substrate. The technical core of this achievement lies in the successful heteroepitaxial growth of

a vertical 2DEG channel structure directly on silicon. This innovative approach overcomes the substantial lattice and thermal mismatch challenges typically associated with foreign substrates, realizing high-quality vertical transport channels on a standard silicon platform.

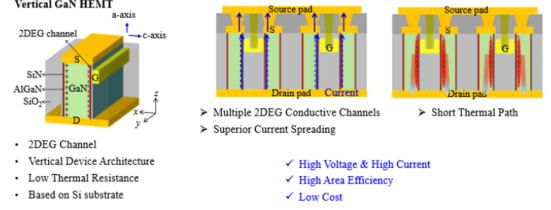


Fig. 8: vGaN HEMT

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From a commercial perspective, this technology fundamentally alters the cost structure of high-power electronics. By effectively decoupling high performance from the prohibitive costs of native GaN substrates, this breakthrough positions vertical GaN as a formidable competitor to incumbent Si-IGBTs and lateral GaN devices. Consequently, this innovation holds the potential to accelerate the adoption of high-efficiency vertical GaN solutions within cost-sensitive mass markets that were previously inaccessible.

B. Hybrid GaN-on-SiC Devices

To synthesize the distinct advantages of wide-bandgap materials, researchers have engineered hybrid device architectures that integrate a Gallium Nitride (GaN) channel layer atop a Silicon Carbide (SiC) substrate. In this configuration, the SiC acts as both

the drain and the drift region, while the GaN layer is utilized specifically for the channel. This heterogenous integration aims to overcome the specific material bottlenecks encountered when using either semiconductor in isolation.

The primary motivation for this hybrid structure is the powerful synergy between the physical properties of the two materials. Thermally, the device exploits SiC's exceptional thermal conductivity—approximately three times that of Silicon and 1.5 times that of GaN—to provide a superior heat dissipation pathway, addressing the thermal management issues often faced by high-power GaN devices. Electrically, the SiC drift layer imparts robust avalanche capability, significantly enhancing device ruggedness, while the GaN top layer contributes its characteristic high electron mobility, ensuring low channel resistance and enabling high-speed switching operations.

Experimental realizations, particularly in the form of hybrid GaN-on-SiC Trench MOSFETs, have successfully demonstrated the efficacy of this design. These devices achieve fully vertical conduction, effectively combining the high-speed potential of GaN with the thermal and structural stability of SiC. Crucially, performance characterization reveals a marked reduction in thermal resistance compared to standard GaN-on-Silicon devices, positioning this hybrid technology as a robust solution for high-power density applications where thermal limitations are a critical constraint.

C. Optically Triggered Power Devices

MIT researchers demonstrated optically controlled vertical GaN FinFETs. By integrating a photosensitive gate, the device can be switched using light. This eliminates the need for electrical gate drivers and galvanic isolation components, significantly reducing the size and EMI susceptibility of high-voltage power modules.

VII. APPLICATIONS AND FUTURE OUTLOOK

The unique properties of v-GaN position it for specific high-power sectors where lateral GaN and SiC face limitations.

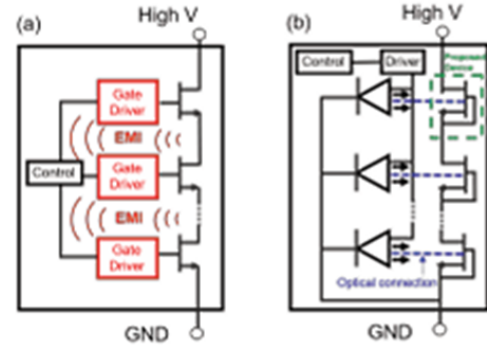


Fig. 9: Optical Trigger

Power Switching Technologies for Various Applications

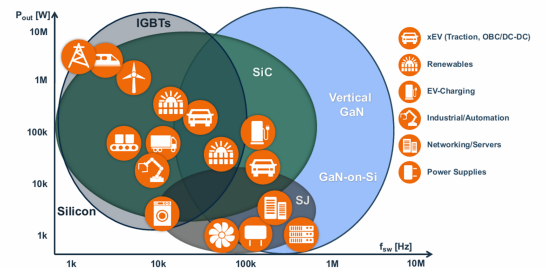


Fig. 10: Applications

A. AI Data Centers

AI processors require massive currents at low voltages (48V to <1V conversion). Vertical GaN devices allow for 3x higher power density in DC-DC converters. By enabling higher switching frequencies, v-GaN reduces the size of passive components (inductors/capacitors), freeing up valuable board space for compute engines.

B. Electric Vehicle (EV) Powertrains

For 800V battery architectures, v-GaN offers an ideal solution for On-Board Chargers (OBC) and traction inverters. ONSEMI has announced 1200V v-GaN devices sampling in 2025, targeting mass production by 2026. The vertical architecture provides the necessary avalanche ruggedness required for automotive safety standards.

C. Renewable Energy Grids

Solar inverters and wind turbines require high-voltage switching. v-GaN devices rated for 1200V–3300V can replace bulky Si-IGBT modules, reducing switching losses and improving system efficiency.

VIII. CONCLUSION

Vertical GaN (v-GaN) power devices have progressed rapidly from conceptual studies to experimentally validated prototypes, marking a significant inflection point in the evolution of wide-bandgap power electronics. By enabling current transport through the bulk drift region rather than along a surface-confined 2DEG channel, vertical architectures fundamentally resolve the intrinsic scaling, electric-field crowding, and reliability limitations encountered in lateral GaN HEMTs. This shift in device topology not only permits breakdown voltage to scale with epitaxial thickness but also facilitates avalanche-capable operation and substantially higher current density, positioning v-GaN as a compelling candidate for high-voltage and high-power applications.

Despite the clear material advantages of GaN, the widespread adoption of vertical devices has historically been constrained by the cost and availability of high-quality native GaN substrates. Recent developments—including the demonstration of vertical GaN-on-Si HEMTs and GaN-on-SiC hybrid structures—provide practical pathways toward reducing substrate cost while maintaining acceptable crystalline quality and thermal performance. In parallel, significant advances in enabling process technologies such as thick-film homoepitaxy, high-activation-energy p-type doping strategies, and low-damage trench etching have expanded the design space for high-performance vertical MOSFETs, CAVETs, and FinFETs.

Edge-termination engineering has similarly advanced, with techniques such as fluorine-ion-implanted termination (FIT), optimized bevel structures, and multi-ring junction extensions enabling devices to approach the theoretical breakdown limits of GaN. Several demonstrations have already achieved voltage ratings exceeding 1.2 kV with competitive specific on-resistance, validating the scalability and robustness of v-GaN architectures.

Looking forward, continued progress in substrate engineering, defect reduction, and manufacturable process integration will be essential to transition v-GaN

from laboratory-scale prototypes to high-volume commercial deployment. With these advancements, vertical GaN is well-positioned to become a foundational technology for next-generation power electronics in data-center infrastructure, electric vehicle powertrains, renewable-energy conversion, and other high-efficiency, high-power-density applications.

IX. ACKNOWLEDGMENT

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